



Lecture 2 : Game design, Methodologies

By Antoine McNabb-Baltar

Game design and game development



What is a game ?

A game is an **activity** or sport involving skill, knowledge, and/or chance, in which you follow fixed rules and try to achieve a specific goal.

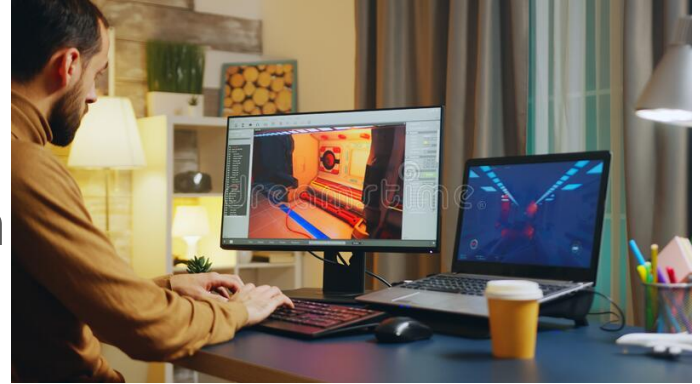
A **game** needs **goals** and **rules** (or mechanics)

Gameplay is the interactive aspects of video game design

- Gameplay involves player interaction with the game, usually for the purpose of entertainment, education or training.

What is game design?

Game design is the process of creating and shaping the **mechanics, systems, and rules** of a game.





Different area of game design

World Design :

- World design is the creation of a backstory, setting, and theme for the game;

System Design:

- System design is the creation of game rules and underlying mathematical patterns.

Content Design :

- Content design is the creation of characters, items, puzzles, and missions. Game Writing Game writing involves writing dialogue, text, and story.

Level Design :

- Level design is the construction of world levels and its features.

User Interface Design

- User interface designer constructs the user interactions and feedback interface, like menus or heads-up displays.



Game design document

A **game design document** (often abbreviated GDD) is a highly **descriptive living design** document of the game design for a video game.

The purpose of a game design document is to **unambiguously** describe the **game's selling** points, target audience, gameplay, art, level design, story, characters, UI, assets, etc.

In short, **every game part requiring development** should be included by the **developer**.

The document is purposely **sectioned** and divided in a way that game developers can **refer** to and **maintain** the relevant parts.



Some good examples of game design documents

<https://www.gamedevs.org/uploads/grand-theft-auto.pdf>

https://drive.google.com/file/d/1nxvdXasP-HsRCt62cHK3wF_plrJpYx5T/view

https://drive.google.com/file/d/1nxvdXasP-HsRCt62cHK3wF_plrJpYx5T/view



Fundamental of game design : Game mechanics

Game mechanics are the **rules** and **procedures** that guide the player **and** the game response to the player's moves or actions. Through the mechanics you create, you define how the game is going to work for the people who play it.

A game mechanic has **3** important parts :

The Discovery :

How you discover the mechanic. How it is taught to player (through level design , through tutorial)

The Motivation :

Why are you using the mechanic ? What is your end goal ?

The Challenge :

How do we test the player proficiency with the mechanic

A quick overview of phase of development

A game project will generally go through 3 stages of development: **Pre-production**, **Production**, **Post-Production**

Pre-production :

Pre-production is about deciding on **key factors** in the pipeline process. Generally, Pre-Production begins by addressing things like what **resources** and **budget** will be required to produce the title.

It will typically involve a **producer** and some **game directors** as well as some **programmers** and **concept artist**.





A quick overview of phase of development

Production : Production is the longest and most important stage of game development. This is where the lion's share of your time and money will be spent, and your game truly starts to come to life.

It is the phase where we truly build the game we set out to do.

It is generally split into 7 deliverable :

Prototype, First Playable , Vertical Slice , Pre-alpha, Alpha, Beta, Gold Master



Production deliverables

Prototype: Initial test of core mechanics, very bare bones

First Playable: Incorporates the visual elements with the core mechanics

Vertical Slice: Essentially a completed sample that can be used for marketing or pitching purposes

Pre-alpha: The content for the title is mostly complete, however this is where permanent choices are made over core elements



Production deliverables

Alpha: Game is “**Feature Complete**” , this refers to the fact that all the main elements of the game are in place and the product is now playable. It may be missing assets and some FX at this point, but this stage is ready for internal user testing

Beta: Content, Assets and core functions are completely in place, this stage of the pipeline is purely optimization based

Gold Master: The Game is finally ready to be published and shipped for the public to enjoy.

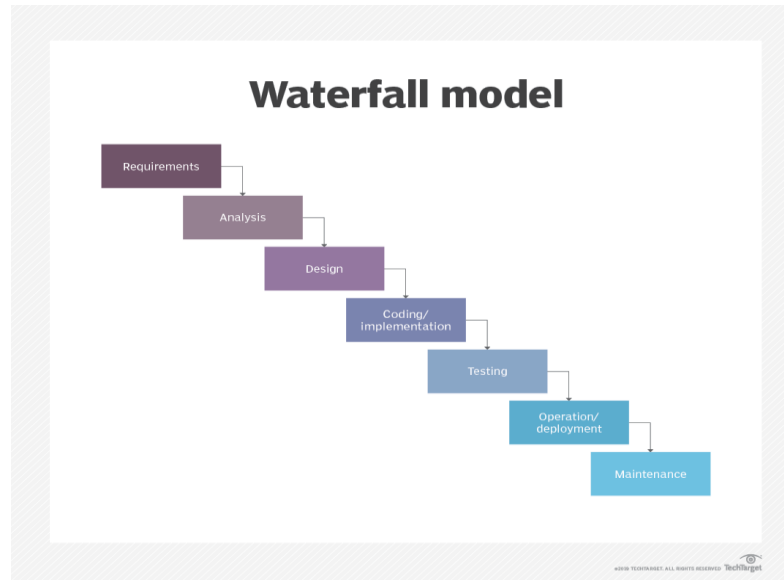


A quick overview of phase of development

Post production : It's a phase of **maintenance**. Part of the resources used for production are reallocate to maintenance, bug fixing, and developing patches to improve the overall game experience. Whilst QA finds 90% of bugs for the developers to address and fix, the **massive player base** of titles will **always** find **bugs** missed in that were missed during the QA process.

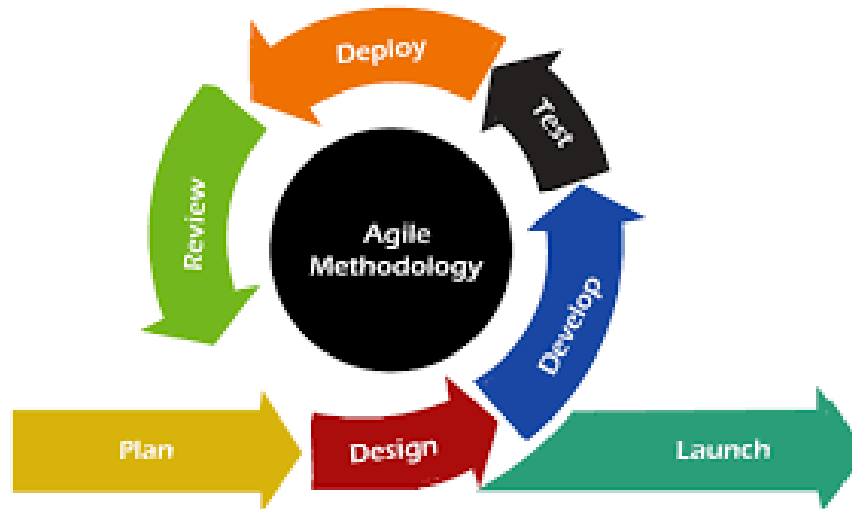
Project management methodologies

Waterfall model :



Project management methodologies

Agile :





Why do we use the agile methodologies in game ?

Game developpement uses the **agile methodology**.

Because it allows teams to quickly adapt to requirements changes/ design changes **without negatively impacting release dates**. Not only that, **Agile** helps reduce technical debt, allows to have a **continuously playable** product and deliver a higher quality product.